

An Evidence-Driven ePolyScat Research Portal

End-to-End Scientific Workflows, Reproducible Inputs, and Verified Computation

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PROJECT ACCEPTANCE
REVIEW
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Scientific Gateway Delivery

PROJECT OBJECTIVE

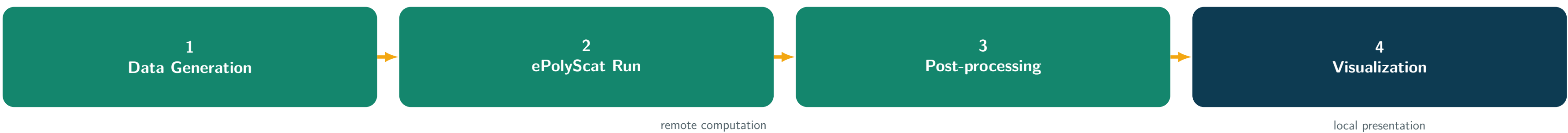
Turn fragmented scientific tools into a traceable, reviewable research workflow.

CONTRIBUTION A · ZIMING ZHANG

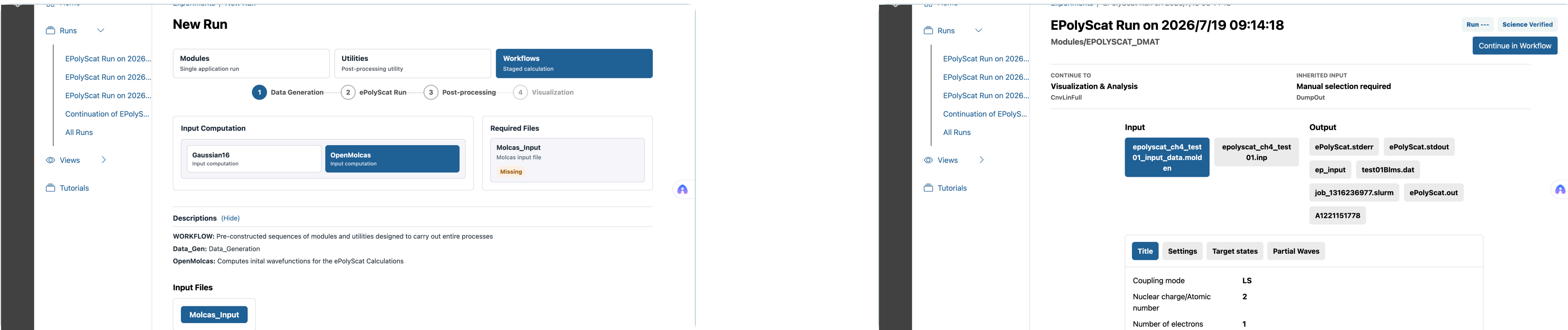
Workflow Portal and Scientific Verification

The implemented portal now exposes a usable, reviewable path from scientific input to verified HPC output and the next workflow stage.

Four-stage research journey



Portal evidence



Left: guided Workflow mode exposes all four stages and required-file readiness. Right: Portal Run 31 shows **Science Verified**, output artifacts, and the continuation contract for Visualization & Analysis.

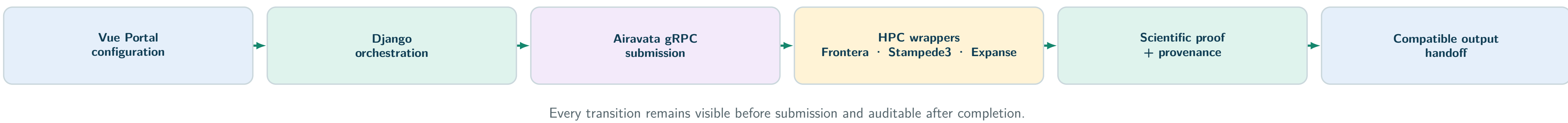
Reproducible inputs

- File View and Table View stay synchronized.
- An ordered, lossless document model preserves repeated records, commands, comments, blank lines, and unknown input.
- Ordered Sequence editing never replaces source text with generated defaults.

Verified execution contracts

- Output manifests carry semantic roles and provenance.
- Continuation requires scientific evidence—not scheduler completion alone.
- Resource-aware wrappers propagate failure and verify application-specific completion markers.

System path: one reviewable chain



Acceptance evidence

296

non-gRPC automated checks passed

3 / 3

HPC wrapper regression suites passed

34 · 67 · 24.9k

implementation commits · changed files · lines added

Portal evidence: Run 23 / Expanse verified OpenMolcas and inherited .scf.molden; Run 31 / Frontera verified ePolyScat and produced test01B1ms.dat.

Remote evidence: Frontera job 7861631 and Stampede3 jobs 3316916 / 3316918 completed with application-specific EDCS or OpenMolcas success markers. Vue production build also passed.

ACCEPTANCE CLAIM

Connect configuration, execution, scientific validation, and stage-to-stage handoff in one portal experience.

CONTRIBUTION B · NGUYEN, THINH M

[Partner Progress Title]

[One sentence: what changed, for whom, and why it matters.]

Problem & method

[Problem and method: summarize the approach in two or three concise points.]

Implementation

[Implementation or experimental contribution: name the system, artifact, or analysis delivered.]

Primary result

PRIMARY RESULT FIGURE
Replace \PartnerVisual with
\includegraphics[width=\linewidth]{...}

[One sentence explaining what the visual proves.]

Validation evidence

✓ [Tests, measurements, comparison, or user evidence supporting the claim.]

Outcome metrics

[Two to four defensible outcome metrics, separated by centered dots.]

How this contribution fits the shared objective

01

Traceable configuration

02

Verified computation

03

Reviewable research output

REPLACEMENT GUIDE Edit the eight \Partner... macros near the top; every labeled area updates automatically.